METRIC

DOD-HDBK-270(SH)
1 June 1983

MILITARY HANDBOOK

METALLIC MATERIALS, MAGNETIC PERMEABILITY AND ELECTRICAL CONDUCTIVITY, CHARACTERISTICS OF (METRIC)

FSC 95GP

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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362

DOD-HDBK-270(SH)

Metallic Materials, Magnetic Permeability and Electrical Conductivity, Characteristics of (Metric).

1. This Military Standardization Handbook was developed by the Naval Sea Systems Command, Department of the Navy, in accordance with established procedures.

2. This publication was approved on 1 June 1983, for printing and inclusion in the military standardization handbook series.

3. This document provides basic and fundamental information on the magnetic permeability and electrical conductivity characteristics of metals and alloys. The handbook is not intended to be referenced in purchase specifications except for informational purposes, nor shall it supersede any specification requirements.

4. Every effort has been made to reflect the latest information on the magnetic permeability and electrical conductivity characteristics of the metals and alloys listed. It is the intent to review this handbook periodically to ensure its completeness and accuracy.

5. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. This handbook provides design engineers and equipment manufacturers general information on magnetic permeability and electrical conductivity characteristics of metals and alloys used in manufacturing equipments used in magnetic sensitive areas in Naval ships. The metals and alloys included are primarily those having low (less than 2.0) relative magnetic permeability in the mill annealed condition. However, other metals and alloys with higher magnetic permeability are also included because their use in equipments is dictated by consideration of other properties which are not always available in low magnetic permeability materials.

2. The values given herein should be considered as "typical", and are intended to show the relative effect of cold-working operations on the characteristics cited.

3. The numerical values for properties listed in this handbook which were derived from specifications, standards, handbooks and other publications, reflect the values in issues in effect on the date of approval of this handbook. Because of revisions or amendments to these documents after publication, the handbook values may differ from those shown in current specifications, standards, and handbooks. When used in acquisition, it should be understood that the governing requirements are those of the documents of the issue in effect on date of invitation for bids or request for proposal.

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1. SCOPE

1.1 Scope. This handbook provides the following technical data and information for metals and alloys (in particular, copper, nickel, and aluminum alloys, and for corrosion-resisting (stainless) steel):

(a) Common name.
(b) Commercial designation.
(c) UNS (unified numbering system) number.
(d) Chemical composition.
(e) Applicable Government and commercial specifications.
(f) Normal and maximum magnetic permeabilities at selected percentages of cold-working.
(g) Electrical conductivity.
(h) Magnetic permeability data source.

1.1.1 Application. The technical data and information included in this handbook is not intended for use in material applications, but will provide a source of information for typical values of magnetic permeability and electrical conductivity of the listed metals and alloys.

1.1.2 Arrangement of handbook. Tables I through III contain data and information for alloys. Tables IV through VII contain data and information for corrosion-resisting (stainless) steels. Table VIII lists magnetic permeability and electrical conductivity data sources. Tables IX, X, and XI contain cross-references of Government and commercial specifications, and UNS numbers to Government specifications.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this handbook to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-596 - Aluminum Alloy Permanent and Semipermanent Mold Castings.
QQ-A-601 - Aluminum Alloy Sand Castings.
QQ-B-626 - Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-637 - Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire and Strip).
QQ-B-639 - Brass, Naval: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-728 - Bronze Manganese; Rod, Shapes, Forgings and Flat Products (Flat Wire, Strip, Sheet, Bar and Plate).
QQ-C-390 - Copper Alloy Castings (Including Cast Bar).
QQ-C-465 - Copper-Aluminum Alloys (Aluminum Bronze) (Copper Alloy Numbers 606, 614, 630, 632M, and 642); Rod, Flat Products with Finished Edges (Flat Wire, Strip, and Bar) Shapes, and Forgings.

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FEDERAL (Continued)

QQ-C-591 - Copper-Silicon, Copper-Zinc Silicon, and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum-Alloy, Wrought.
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
QQ-S-763 - Steel Bars, Wires, Shapes, and Forgings, Corrosion Resisting.
QQ-S-766 - Steel Plates, Sheets, and Strip-Corrosion Resisting.
QQ-W-423 - Wire, Steel, Corrosion-Resisting.

MILITARY

MIL-S-862 - Steel Bars, Corrosion Resisting, and Steel Billets, Corrosion Resisting: Reforging Application.
MIL-S-5059 - Steel, Corrosion-Resistant (18-8), Plate, Sheet and Strip.
MIL-T-5695 - Tubing, Steel, Corrosion-Resistant (304), Cold Drawn.
MIL-W-6712 - Wire; Metallizing.
MIL-T-8231 - Tubes: Copper-Silicon-Bronze, Seamless, for Aircraft.
MIL-C-11866 - Castings, Precision, Non-ferrous.
MIL-T-15005 - Tubes, 70-30 and 90-10 Copper-Nickel Alloy, Condenser and Heat Exchanger.
MIL-C-15345 - Castings, Nonferrous, Centrifugal.
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MIL-B-16166 - Bronze, Aluminum: Forgings, Heat-Treated.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-N-17163 - Nickel-Copper Alloy, Wrought; (55-60 Percent Nickel) Low Permeability.
MIL-S-17759 - Steel Bar, Billet, Forging and Wire-Chromium-Nickel-Phosphorus and Chromium-Nickel-Manganese-Phosphorus Alloy, Low Magnetic Permeability.
MIL-S-17996 - Steel Sheet and Other Wrought Forms, Corrosion Resisting, Austenitic (Low Magnetic Permeability), Chromium-Manganese-Nickel.
MIL-B-21230 - Bronze, Nickel Aluminum and Manganese-Nickel Aluminum: Castings, Ship Propeller Application.
MIL-C-22087 - Copper Alloy Investment Castings.
MIL-C-22229 - Copper Base Alloy Castings (For Pressure Tight Applications).
MIL-C-24111 - Chrominum-Nickel Alloy Steel Bars, Rods and Forgings, Corrosion Resistant, (Precipitation Hardening).
MIL-S-25043 - Steel Plate, Sheet, and Strip, 17-7 Ph, Corrosion-Resistant, Precipitation Hardening.
MIL-S-81591 - Steel, Investment Castings, Carbon and Corrosion-Resisting.

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STANDARDS

MILITARY

MIL-STD-288 - Inspection Procedure for Determining the Magnetic Permeability of Wrought Austenitic Steels.
DOD-STD-2141 - Definitions and Systems of Units, Magnetic Silencing (Metric).

HANDBOOKS

MILITARY

MIL-HDBK-694 - Aluminum and Aluminum Alloys.
MIL-HDBK-698 - Copper and Copper Alloys.
MIL-HDBK-723 - Steel and Iron Wrought Products.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this handbook to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
A 177 - High-Strength Stainless and Heat-Resisting Chromium-Nickel Steel Sheet and Strip.
A 213 - Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes.
A 240 - Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Fusion-Welded Unfired Pressure Vessels.
A 249 - Welded Austenitic Steel Boiler, Superheater, Heat-Exchanger, and Condenser Tubes.
A 262 - Detecting Susceptibility to Intergranular Attack in Austenitic Stainless Steels.
A 269 - Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
A 276 - Stainless and Heat-Resisting Steel Bars and Shapes.
A 289 - Alloy Steel Forgings for Nonmagnetic Retaining Rings for Generators.
A 297 - Heat-Resistant Iron-Chromium and Iron-Chromium-Nickel Alloy Castings for General Application.
A 312 - Seamless and Welded Austenitic Stainless Steel Pipe.
A 403 - Wrought Austenitic Stainless Steel Piping Fittings.
A 412 - Stainless and Heat-Resisting Chromium-Nickel-Manganese Steel Plate, Sheet, and Strip.

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A 453 - Bolting Materials, High-Temperature, 50 to 120 KSI Yield Strength, with Expansion Coefficients Comparable to Austenitic Steels.
A 479 - Stainless and Heat-Resisting Steel Wire, Bars and Shapes for Use in Boilers and Other Pressure Vessels.
A 493 - Stainless and Heat-Resisting Steel for Cold Heading and Cold Forging - Bar and Wire.
A 498 - Seamless and Welded Carbon, Ferritic, and Austenitic Alloy Steel Heat-Exchanger Tubes with Integral Fins.
A 511 - Seamless Stainless Steel Mechanical Tubing.
A 554 - Welded Stainless Steel Mechanical Tubing.
A 564 - Hot-Rolled and Cold-Finished Age-Hardening Stainless and Heat-Resisting Steel Bars, Wire, and Shapes.
A 632 - Seamless and Welded Austenitic Stainless Steel Tubing (Small-Diameter) for General Service.
A 638 - Precipitation Hardening Iron Base Superalloy Bars, Forgings, and Forging Stock for High-Temperature Service.
A 651 - Stainless Steel Water-DWV Tube.
A 666 - Austenitic Stainless Steel, Sheet, Strip, Plate, and Flat Bar for Structural Applications.
A 688 - Welded Austenitic Stainless Steel Feedwater Heater Tubes.
B 16 - Free-Cutting Brass Rod, Bar, and Shapes for Use in Screw Machines.
B 21 - Naval Brass Rod, Bar and Shapes.
B 26 - Aluminum-Alloy Sand Castings.
B 30 - Copper-Base Alloys in Ingot Form.
B 61 - Steam or Valve Bronze Castings.
B 96 - Copper-Silicon Alloy Plate Sheet, Strip, and Rolled Bar for General Purposes.
B 98 - Copper-Silicon Alloy Rod, Bar, and Shapes.
B 100 - Rolled Copper-Alloy Bearing and Expansion Plates and Sheets for Bridge and other Structural Uses.
B 108 - Aluminum-Alloy Permanent Mold Castings.
B 111 - Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock.
B 121 - Leaded Brass Plate, Sheet, Strip, and Rolled Bar.
B 122 - Copper-Nickel-Tin Alloy, Copper-Nickel-Zinc Alloy (Nickel Silver), and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar.
B 124 - Copper and Copper-Alloy Forging Rod, Bar, and Shapes.
B 138 - Manganese Bronze Rod, Bar, and Shapes.
B 148 - Aluminum-Bronze Sand Castings.
B 150 - Aluminum Bronze Rod, Bar, and Shapes.
B 151 - Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Rod and Bar.
B 164 - Nickel-Copper Alloy Rod and Bar.
B 171 - Copper-Alloy Condenser Tube Plates.
B 174 - Bunch-Stranded Copper Conductors for Electrical Conductors.
B 193 - Resistivity of Electrical Conductor Materials.

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B 206 - Copper-Nickel-Zinc, Alloy (Nickel Silver) Wire and Copper-Nickel Alloy Wire.
B 271 - Copper-Base Alloy Centrifugal Castings.
B 283 - Copper and Copper-Alloy Die Forgings (Hot-Pressed).
B 315 - Seamless Copper Alloy Pipe and Tube.
B 359 - Copper and Copper-Alloy Seamless Condenser and Heat Exchanger Tubes with Integral Fins.
B 395 - U-Bend Seamless Copper and Copper Alloy Heat Exchanger and Condenser Tubes.
B 402 - Copper-Nickel Alloy Plate and Sheet for Pressure Vessels.
B 466 - Seamless Copper-Nickel Pipe and Tube.
B 467 - Welded Copper-Nickel Pipe.
B 505 - Copper-Base Alloy Continuous Castings.
B 543 - Welded Copper and Copper-Alloy Heat Exchanger Tube.
B 552 - Seamless and Welded Copper/Nickel Tubes for Water Desalting Plants.
B 584 - Copper Alloy Sand Castings for General Applications.
B 608 - Welded Copper-Alloy Pipe.
B 637 - Precipitation Hardening Nickel Alloy Bars, Forgings, and Forging Stock for High-Temperature Service.
B 639 - Precipitation Hardening Cobalt-Containing Alloy Rod, Bar, Forgings, and Forging Stock for High-Temperature Service.
E 527 - Numbering Metals and Alloys (UNS).
F 467 - Nonferrous Nuts for General Use.
F 468 - Nonferrous Bolts, Hex Cap Screws, and Studs for General Use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. DEFINITIONS

3.1 General. The general magnetic terms used in this handbook are defined in DOD-STD-2141. The following definitions apply specifically to the use of the terms in this handbook.

3.1.1 Cold-work. Cold-work refers to the metalworking operations carried on at or near room temperature which result in plastic deformation of the metal. It may be applied to most of the common metals. Since no recovery occurs in most cases at this temperature, the properties of the metal are altered in the direction of increasing strength and brittleness throughout the working process. Depending upon the chemical composition of the metal and the heat treatment used, the magnetic permeability of the metal may increase through cold-working. Consequently, a nonmagnetic metal with no cold-work may become magnetic after cold-working.

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3.1.2 Maximum magnetic permeability. Maximum magnetic permeability refers to the largest value of normal magnetic permeability obtained by varying the amplitude of the magnetic field strength in which the metal is placed.

3.1.3 Normal magnetic permeability. Normal magnetic permeability refers to the relative magnetic permeability of a metal subjected to a field strength equal to the earth's magnetic field (approximately 50,000 nanotesla). Unless otherwise specified, all normal magnetic permeabilities are measured at room temperature (approximately 24 degrees Celsius (°C)).

3.1.4 Percent cold-work. Percent cold-work refers to the percentage reduction in metal thickness when cold-worked. Depending upon the composition of the metal and the heat treatment, the normal magnetic permeability of a metal may vary significantly with cold-working.

3.1.5 Unified numbering system (UNS). UNS refers to an alphanumeric code established by the technical societies (see ASTM E 527), for each family of metals and alloys intended to identify metals and alloys that are in regular production and use. The alphanumeric code is a simple expansion of the former standard designation system accomplished by the addition of a prefix (letter) and suffix (numbers).

3.1.6 Acronyms. Acronyms are defined as follows:

AA - Aluminum Association
AMS - Aeronautical Material Specification
ASTM - American Society for Testing and Materials
EES - Engineering Experiment Station (Now NSRDC-Annapolis)
IACS - International Annealed Copper Standards
NSRDC - Naval Ship Research and Development Center
SAE - Society of Automotive Engineers

3.1.7 Chemical symbols. Chemical symbols used in stating the chemical composition of the various materials are as follows:

Al - aluminum
Sb - antimony
C - carbon
Cr - chromium
Cu - copper
Fe - iron
Pb - lead
Mg - magnesium
Mn - manganese
Mo - molybdenum
Nb - niobium
Ni - nickel
N - nitrogen
P - phosphorus
Si - silicon
S - sulphur
Ta - tantalum
Sn - tin
Ti - titanium
W - tungsten
Zn - zinc

3.1.8 Column headings in tables. Column headings in tables are as follows:

(a) Common name - the common name(s) by which the material is generally known.
(b) Commercial designation - the official designation(s) used by industry and the Government.
(c) UNS No. - the unified numbering system code identifier (see 3.1.5).

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(d) Chemical composition - the chemical compositions conform to the referenced Government specifications (see 5.1.1).

(e) Specifications -

(1) Government - federal and military specifications which cover the material.

(2) Commercial - technical society specifications which cover the material.

(f) Percent cold-work - the amount (in percent) of cold-working applied to the material (see 3.1.1, 3.1.4 and 5.1.2). If unknown, the abbreviation UNK is used.

(g) Magnetic permeability -

(1) Normal - the normal relative magnetic permeability (see 3.1.3), in pure numbers, associated with the amount of cold-work in (f) above.

(2) Maximum - the maximum relative magnetic permeability (see 3.1.2), in pure numbers, associated with the amount of cold-work in (f) above.

(h) Electrical conductivity - the electrical conductivity in percent international annealed copper standards (IACS), relative to the amount of cold-work in (f) above.

(i) Data sources - the sources of magnetic and electrical data are identified by alphanumeric code. The alphabetical notation in parentheses identifies the type of data; (e) for electrical and (m) for magnetic. The numeric notation refers to the source which is identified in table VIII by a corresponding numeric entry. For example: 2(e) identifies the type of data as electrical and the data source as the entry opposite the number 2 in table VIII. The absence of a parenthetical expression after a number indicates the data source supplied both electrical and magnetic data.

4. GENERAL REQUIREMENTS

4.1 Metals - general.

4.1.1 Chemical composition. The chemical compositions shown in the tables conform to the referenced Government specifications. The composition may vary slightly from those in the referenced commercial specification and Government specifications where multiple listings are shown. When no Government specifications are listed, the composition is the referenced commercial specification.

4.1.2 Magnetic permeability and electrical conductivity. The relative magnetic permeability and electrical conductivity for a given metal or alloy will vary depending upon the mechanical and heat treatments to which it is subjected. Even though some of these treatments are identified, the magnetic permeabilities and electrical conductivities listed are to be considered typical for the metal or alloy. For some metals and alloys, multiple sets of data for percentages of cold-working are provided to illustrate the relative effects that the different cold-working operations have on the magnetic permeability of the metal when subjected to different heat treatments.

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5. DETAILED REQUIREMENTS

5.1 Metal alloys.

5.1.1 Copper alloys. Table I contains technical data and information on copper alloys. Further data and information on copper alloys may be found in MIL-HDBK-698.

5.1.2 Nickel alloys. Table II contains technical data and information on nickel alloys. The ferromagnetic properties of the nickel alloys are highly susceptible to change due to variations in chemical composition, or to mechanical and heat treatments.

5.1.3 Aluminum alloys. Table III contains technical data and information on aluminum alloys. Further data and information on aluminum alloys may be found in MIL-HDBK-694.

5.2 Corrosion resistant (stainless) steels.

5.2.1 Wrought austenitic corrosion resistant steels. Table IV contains technical data and information on wrought austenitic corrosion resistant steels. Further data and information on wrought steel products may be found in MIL-HDBK-723 and in MIL-STD-288 which classifies wrought austenitic steels. Where these steels are listed in table IV, their classifications are noted.

5.2.2 Precipitation hardening corrosion resistant steels. Table V contains technical data and information on precipitation hardening corrosion resistant steels.

5.2.3 Cast austenitic corrosion resistant steels. Table VI contains technical data and information on cast austenitic corrosion resistant steels, cast irons and welds.

5.2.4 Austenitic manganese steels. Table VII contains technical data and information on austenitic manganese steels.

6. NOTES

6.1 Data sources. The sources of the magnetic permeability and electrical conductivity data are given in table VIII. Each source has been assigned a number which relates to the numeric part of the code in the data sources column of tables I through VII (see 3.1.8(i)).

6.2 Cross-indexes. Tables IX, X and XI cross-index the following:

(a) Table IX - Government specifications to commercial specifications and includes metal/alloy common names and UNS numbers.

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(b) Table X - Commercial to Government specifications.
(c) Table XI - UNS numbers to Government specifications.

For further cross-indexing of metals and alloys, refer to ASTM DS-56A.

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TABLE I. Copper alloys.

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Brass, free cutting	SAE CA360	C36000	60.0 - 63.0 Cu 2.5 - 3.7 Pb 0.35 Fe 0.50 others rem Zn	QQ-B-626 alloy 360	ASTM B 16	UNK	1.04		26	1(m) 4(e)	
Naval brass, uninhibited	SAE CA464	C46400	59.0 - 62.0 Cu 0.5 - 1.0 Sn 0.20 Pb 0.10 Fe 0.10 others rem Zn	QQ-B-637 alloy 464 QQ-B-639	ASTM B 21 B 124 B 174 B 283	UNK	1.004		26	1(m) 4(e)	
Red brass, leaded	85-5-5-5 or no. 1 composition SAE CA836	C83600	84.0 - 86.0 Cu 4.0 - 6.0 Sn 4.0 - 6.0 Pb 4.0 - 6.0 Zn 0.35 Fe 1.0 Ni 0.05 P	MIL-C-22229 alloy 836 MIL-C-22087 alloy 836 MIL-C-15345 alloy 1	ASTM B 30 B 271 B 505 B 584	UNK	1.02		15	1(m) 4(e)	
Aluminum bronze	Grade A AMPCO 12	C95200	86.0 min Cu 8.5 - 9.5 Al 2.5 - 4.0 Fe 0.50 others	QQ-C-390 alloy 952	ASTM B 148	0	1.19		11	1(m) 4(e)	

See footnotes at end of table.

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TABLE I. Copper alloys. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Aluminum bronze	Grade C AMPCO 18	C95400	83.0 min Cu 10.0 - 11.5 Al 3.0 - 5.0 Fe 0.50 Mn 2.5 Ni 0.50 others	QQ-C-390 alloy 954	ASTM B 148	UNK	1.15	1.52	13	1(m) 4(e)	
Aluminum manganese bronze	---	C95700	71.0 Cu 7.0 - 8.5 Al 2.0 - 4.0 Fe 1.5 - 3.0 Ni 11.0 - 14.0 Mn 0.03 Pb 0.10 Si 0.50 others	QQ-C-390 alloy 957	ASTM B 30 B 148	0	1.22		3.1	1(m) 4(e)	
Aluminum nickel bronze	AMPCO 45	C63000	78.0 - 85.0 Cu 9.0 - 11.0 Al 2.0 - 4.0 Fe 4.0 - 5.5 Ni 1.5 Mn 0.20 Sn 0.30 Zn 0.50 others	QQ-C-465 alloy 630 MIL-B-16166	ASTM B 150 B 283 B 171 B 124	0	1.04	1.05	9	1(m) 4(e)	
Aluminum nickel bronze	---	C95800	78.0 Cu 8.5 - 9.5 Al 3.0 - 5.0 Fe 4.0 - 5.5 Ni 3.5 Mn 0.02 Pb 0.10 Si 0.50 others	QQ-C-390 alloy 958 MIL-C-15345 alloy 28 MIL-B-21230 alloy 1	ASTM B 30 B 148 B 271 B 505	0	1.22		7.1	1(m) 4(e)	

See footnotes at end of table.

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TABLE I. Copper alloys. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Tin nickel bronze	---	C94700	86.0 - 89.0 Cu 4.7 - 6.0 Sn 0.08 Pb 1.3 - 2.5 Zn 4.5 - 6.0 Ni 0.20 Fe 0.05 P 0.20 Mn 0.10 Sb 0.25 others	QQ-C-390 alloy 947	ASTM B 30 B 505 B 584	0		1.00		1(m)	
Tin nickel bronze	---	C94800	85.0 - 89.0 Cu 4.7 - 6.0 Sn 0.30 - 0.90 Pb 1.3 - 2.5 Zn 4.5 - 6.0 Ni 0.20 Fe 0.05 P 0.20 Mn 0.10 Sb 0.25 others	QQ-C-390 alloy 948	ASTM B 505 B 584	0		1.00	12	1(m) 4(e)	
High-strength yellow brass	High strength manganese bronze SAE CA862	C86200	60.0 - 66.00 Cu 0.20 Sn 0.20 Pb 2.0 - 4.0 Fe 3.0 - 4.9 Al 2.5 - 5.0 Mn rem Zn	MIL-C-11866 CMPSN 20 MIL-C-22229 CMPSN 862	ASTM B 30 B 271 B 505 B 584	UNK	1.27		7.5	1(m) 4(e)	

See footnotes at end of table.

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TABLE I. Copper alloys. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Tin bronze	88-8-0-4 or modified tin SAE CA903	C90300	86.0 - 89.0 Cu 7.5 - 9.0 Sn 0.30 Pb 3.0 - 5.0 Zn 0.15 Fe 1.0 Ni 0.05 P	MIL-C-15345 alloy 8 MIL-C-22229 alloy 903	ASTM B 30 B 271 B 505 B 584	UNK	1.004		12	1(m) 4(e)	
Manganese bronze, case						0	1.09			1(m)	
Manganese bronze	AMPCO 62					0	1.14			1(m)	
Manganese bronze	No. 1 manganese SAE CA43	C86500	55.0 - 60.0 Cu 1.0 Sn 0.40 Pb 0.40 - 2.0 Fe 0.50 Ni 0.50 - 1.5 Al 1.5 Mn rem Zn	MIL-C-15345 alloy 4 MIL-C-22087 alloy 865 MIL-C-22229 alloy 865	ASTM B 30 B 271 B 584	UNK	1.10		22	1(m) 4(e)	
Leaded tin bronze	AMPCO 71	C92200				0	1.02		14.3	1(m) 4(e)	
Manganese bronze, wrought	Manganese bronze A	C67500	63 - 68 Cu 0.50 Sn 2.0 - 4.0 Fe 2.5 - 5.0 Mn 3.0 - 6.0 Al 0.20 Pb 0.10 others rem Zn	QQ-B-728 class A	ASTM B 124 B 138 B 283	0	1.09		24	1(m) 4(e)	

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TABLE I. Copper alloys. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Leaded tin bronze	Steam or valve bronze-Navy "M" SAE CA922	C92200	86.0 - 90.0 Cu 5.5 - 6.5 Sn 1.0 - 2.0 Pb 3.0 - 5.0 Zn 0.25 Fe 1.0 Ni 0.05 P	MIL-C-15345 alloy 9	ASTM B 30 B 61 B 271 B 505 B 584	UNK	1.02		14.3	1(m) 4(e)	
70-30 copper-nickel alloy	---	C71500	65.0 Cu 29.0 - 32.0 Ni 1.0 Zn 0.40 - 0.70 Fe 0.05 Pb 1.0 Mn 0.02 P 0.02 S	MIL-T-15005 MIL-C-15726 CMPSN 70-30 MIL-T-16420 CMPSN 70-30	ASTM B 111 B 121 B 122 B 151 B 171 B 359 B 395 B 402 B 466 B 467 B 543 B 552 B 608	0		1.00	4.6	1(m) 1(e)	
80-20 copper-nickel alloy	---	C71000	74.0 Cu 19.0 - 23.0 Ni 0.05 Pb 0.50 - 1.0 Fe 1.0 Zn 1.0 Mn	NONE	ASTM B 111 B 122 B 206 B 359 B 395 B 466	0	1.97-1.84			1(m)	

See footnotes at end of table.

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TABLE I. Copper alloys. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Copper-silicon alloy	High silicon bronze A	C65500	94.8 Cu 2.8 - 3.8 Si 1.5 Mn 1.5 Zn 0.80 Fe 0.60 Ni 0.05 Pb 0.50 others	QQ-C-591 alloy 655 MIL-T-8231	ASTM B 96 B 98 B 100 B 124 B 283 B 315	0	1.02		7.0	1(m) 4(e)	
Copper-silicon alloy	Silicon brass	C87400	99.5 Cu 12.0 - 16.0 Zn 1.0 Pb 2.5 - 4.0 Si	QQ-C-390 alloy 874	ASTM B 30 B 271 B 584	0		1.10	6.7	1(m) 4(e)	
Copper-silicon-manganese alloy						0		1.00		1(m)	

1/ Unless otherwise indicated, values shown are maximum permitted.
2/ Percent IACS calculated on a volume basis in accordance with ASTM B 193 from data provided in the indicated data sources.
3/ See 3.1.8(i) for explanation of code.

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TABLE II. Nickel alloys.

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Nickel-copper-aluminum alloy (K-Monel)	Monel alloy 502	N05502	63.0 - 70.0 Ni 2.3 - 3.15 Al 0.25 C 2.0 Fe 1.5 Mn 0.50 Si 0.35 - 0.85 Ti 0.01 S rem Cu 0.02 Zn 0.006 Pb 0.006 Sn	QQ-N-286 class A	NONE	0		1.02	2.8	1(m) 5(e)	
Nickel-copper-alloy (R-Monel)	Class B (Monel alloy R-405)	N04405	63.0 - 70.0 Ni 2.5 Fe 0.50 Al 2.0 Mn 0.3 C 0.5 Si 0.025-0.060 S 0.006 Pb 0.006 Sn 0.02 Zn 0.02 P rem Cu	QQ-N-281 class B	ASTM B 164 F 467 F 468	UNK UNK	4/46.5 4/16.8		3.6	1(m) 8(e)	
Nickel-copper-silicon alloy (S-Monel)			60 Ni 27 - 31 Cu 3.5 - 4.5 Si 2.5 Fe 1.5 Mn 0.25 C 0.50 Al	QQ-N-288 composition D	NONE	0	1.02		2.7	1(m) 5(e)	

See footnotes at end of table.

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TABLE II. Nickel alloys. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Monel	Monel 402 (formerly 326 Monel)		55 - 60 Ni 1.00 Fe 1.25 - 2.25 Mn 3.0 C 0.75 Si 0.024 S rem Cu	MIL-N-17163	NONE	0	5/1.03			1(m)	MIL-N-1716 specifies permeability of 1.1 max at -2.8°C max and field strength of 40,000 nanoteslas

1/ Unless otherwise indicated, values shown are maximum permitted.
2/ Percent IACS calculated on a volume basis in accordance with ASTM B 193 from data provided in the indicated data sources.
3/ See 3.1.8(i) for explanation of code.
4/ Measured in 20,000 nanotesla field.
5/ Permeability may be inversely related to temperature; depending upon the heat treatment, may be magnetic at lower temperatures in a 50,000 nanotesla field.

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TABLE III. Aluminum alloys.

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Aluminum	5 percent silicon Copper-magnesium alloy AA 355.0	A03550	4.5 - 5.5 Si 0.60 Fe 1.0 - 1.5 Cu 0.50 Mn 0.40 - 0.60 Mg 0.25 Cr 0.35 Zn 0.25 Ti 0.15 others rem Al	QQ-A-596 alloy 355 QQ-A-601 alloy 355.0 des T7 MIL-C-11866	ASTM B 26 B 108 grade SC51A	0	1.02		36-43	1(m) 3(e)	
Aluminum	213 (formerly 113) SAE 33		0.6 - 8.0 Cu 1.4 Fe 1.0 - 4.0 Si 0.10 Mg 0.6 Mn 0.35 Ni 2.5 Zn 0.25 Ti 0.50 others rem Al	QQ-A-596 alloy 113	ASTM B 26 grade CS72A	0	1.02		30	1(m) 3(e)	

1/ Unless otherwise indicated, values shown are maximum permitted.
2/ Percent IACS calculated in accordance with ASTM B 193 on a volume basis from data provided in the indicated data sources.
3/ See 3.1.8(i) for explanation of code.

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TABLE IV. Wrought austenitic corrosion resistant steels.

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Chromium-manganese-nickel corrosion resisting (stainless) steel	Type 201	S20100	0.15 C 5.5 - 7.5 Mn 1.0 Si 3.5 - 5.5 Ni 16.0 - 18.0 Cr 0.06 P 0.03 S 0.025 N	4/ QQ-S-766 class 201 5/ MIL-S-17996 class 1	ASTM A 276 A 412 A 666	6/ 0 10 30 50 65		1.005 1.047 5.886 11.72 19.20	2.1 - 2.5	2	Magnetic after moderate cold work
						6/ 0 10 30 50 65		1.035 1.501 16.76 27.44 29.78	2.0		
Chromium-manganese-nickel corrosion resisting (stainless) steel	Type 202	S20200	0.15 C 7.5 - 10.0 Mn 1.0 Si 4.0 - 6.0 Ni 17.0 - 19.0 Cr 0.06 P 0.03 S 0.25 N	4/ QQ-S-763 class 202 4/ QQ-S-766 class 202	ASTM A 276 A 412 A 666	6/ 0 10 30 50 65		1.020 1.037 1.132 5.493 3.404	2.3 - 2.5	2	Magnetic after heavy cold work
						6/ 0 10 30 50 65		1.002 1.002 1.021 1.226 1.707	2.2 - 2.5		
Chromium-manganese-nickel corrosion resisting (stainless) steel	Type 205	S20500	0.12- 0.25 C 14.0- 15.5 Mn 0.03 P 0.03 S 0.50 Si 16.5 -18.0 Cr 1.0 - 1.75 Ni 0.32- 0.40 N	NONE	NONE	6/ 0 10 30 50 65		1.002 1.002 1.003 1.003 1.004	2.2 - 2.3	2	

See footnotes at end of table.

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TABLE IV. Wrought austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Chromium-nickel-corrosion resisting (stainless) steel	Type 301	S30100	0.15 C 2.0 Mn 1.0 Si 6.0 min Ni 16.0 min Cr 0.045 P 0.035 S	4/ QQ-S-766 class 301 MIL-S-5059 type D MIL-T-5695	ASTM A 167 A 177 A 554 grade MT301	6/ 0 10 30 50 65		1.011 1.015 1.698 5.721 12.29	1.7 - 2.5	2	Magnetic after moderate cold work
						6/ 0 10 30 50 65		1.036 1.717 16.05 37.16 45.11	2.2 - 2.5		
						6/ 0 10 30 50 65		1.029 1.630 17.85 37.89 40.77	1.9 - 2.6		
Chromium-nickel-corrosion resisting (stainless) steel	Type 302	S30200	0.15 C 2.0 Mn 0.045 P 0.03 S 1.0 Si 17.0 - 19.0 Cr 8.0 - 10.0 Ni	4/ QQ-S-763 class 302 4/ QQ-S-766 class 302 QQ-W-423 MIL-S-862 MIL-W-6712	ASTM A 167 A 240 A 276 A 479 A 493 A 511 grade MT302 A 554 grade MT302 A 666	6/ 0 10 30 50 65		1.005 1.079 1.144 5.520 11.77	2.2 - 2.5	2	Magnetic after heavy cold work Class 1 in accordance with MIL-STD-28
						6/ 0 10 30 50 65		1.004 1.039 1.414 3.214 7.205	2.2 - 2.4		
Stainless steel	Type 303	S30300	0.15 C 2.0 Mn 0.20 P 0.15 min S 1.0 Si 17.0 - 19.0 Cr 8.0 - 10.0 Ni 0.60 Mo			0 UNK	1.01 1.10			1(m)	

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TABLE IV. Wrought austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Stainless steel	Type 304	S30400	0.08 C 2.0 Mn 1.0 Si 8.00- 12.0 Ni 18.0 - 20.0 Cr 0.045 P 0.030 S	4/ QQ-S-763 class 304 4/ QQ-S-766 class 304	ASTM A 213 grade TP304 A 240 A 249 grade TP304 A 269 A 269 grade TP304 A 276 A 312 grade TP304 A 403 grade TP304 A 479 A 493 A 498 grade TP304 A 511 grade MT304 A 554 grade MT304 A 632 grade TP304 A 666 A 688 grade TP304	6/ 0 10 30 50 65 6/ 0 10 30 50 65		1.083 1.638 4.762 10.01 19.35 1.015 1.064 3.235 8.480 14.41	2.2 - 2.4 2.3 - 2.5	2	Magnetic after heavy cold work Class 1 in accordance with MIL-STD-288

See footnotes at end of table.

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TABLE IV. Wrought austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Stainless steel	Type 305	S30500	0.12 C 2.0 Mn 1.0 Si 10.0 - 13.0 Ni 17.0 - 19.0 Cr 0.045 P 0.030 S	4/ QQ-S-763 class 305 4/ QQ-S-766 class 305	ASTM A 167 A 249 grade TP305 A 276 A 493 A 498 grade TP305 A 511 grade MT305 A 554 grade MT305	6/ 0 10 30 50 65		1.002 1.003 1.004 1.008 1.032	2.3	2	May be slightly magnetic after heavy cold work
Stainless steel	Type 309	S30900	0.20 C 2.0 Mn 1.0 Si 12.0 - 15.0 Ni 22.0 - 24.0 Cr 0.045 P 0.030 S	4/ QQ-S-763 class 309 4/ QQ-S-766 class 309	ASTM A 249 grade TP309 A 276 A 312 grade TP309 A 403 grade TP309 A 498 grade TP309 A 511 grade MT309 A 554 grade MT309	UNK	3.70		2.2	1(m) 2(e)	

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TABLE IV. Wrought austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Stainless steel	Type 310	S31000	0.25 C 2.0 Mn 1.5 Si 19.0 - 22.0 Ni 24.0 - 26.0 Cr 0.045 P 0.030 S	4/ QQ-S-763 class 310 4/ QQ-S-766 class 310	ASTM A 167 A 213 grade TP310 A 249 grade TP310 A 276 A 312 grade TP310 A 403 grade TP310 A 498 grade TP310 A 511 grade MT310 A 554 grade MT310 A 632 grade TP310	6/ 0 10 30 50 65		1.014 1.009 1.012 1.018 1.022	2.0	2	Class 1 in accordance with MIL-STD-288

See footnotes at end of table.

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TABLE IV. Wrought austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Stainless steel	Type 316	S31600	0.08 C 2.0 Mn 1.0 Si 10.0 - 14.0 Ni 16.0 - 18.0 Cr 0.045 P 0.030 S 2.0 - 3.0 Mo	4/ QQ-S-763 class 316 4/ QQ-S-766 class 316	ASTM A 167 A 213 grade TP316 A 240 A 249 grade TP316 A 262 A 269 grade TP316 A 276 A 312 grade TP316 A 403 grade TP316 A 479 A 493 A 498 grade TP316 A 511 grade MT316 A 554 grade MT316 A 632 grade MT316 A 651 grade TP316 A 666 A 688	6/ 0 10 30 50 60		1.010 1.003 1.003 1.004 1.006	2.2 - 2.3	2	May be magnetic heavy cold work Class 2 in accordance with MIL-STD-288

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TABLE IV. Wrought austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Stainless steel	Type 321	S32100	0.08 C 2.0 Mn 1.0 Si 9.0 - 12.0 Ni 17.0 - 19.0 Cr 0.045 P 0.030 S 5 X C, min Ti	4/ QQ-S-766 class 321 4/ QQ-S-766 class 321	ASTM A 167 A 213 grade TP321 A 240 A 249 grade TP321 A 262 A 269 grade TP321 A 276 A 312 grade TP321 A 403 grade TP321 A 479 A 498 grade TP321 A 511 grade MT321 A 554 grade MT321 A 632 grade TP321	6/ 0 10 30 50 65		1.006 1.008 1.033 1.380 2.493	2.3 - 2.4	2	Slightly magnetic after heavy cold work

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TABLE IV. Wrought austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Stainless steel	Type 347	S34700	0.08 C 2.0 Mn 1.0 Si 9.0 - 13.0 Ni 17.0 - 19.0 Cr 0.045 P 0.030 S 10xC, min Nb+Ta	4/ QQ-S-763 class 347 4/ QQ-S-766 class 347	ASTM A 167 A 213 grade TP347 A 240 A 249 grade TP304 A 262 A 269 grade TP304 A 312 grade TP347 A 403 grade TP347 A 479 A 493 A 498 grade TP347 A 511 grade MT347 A 554 grade MT347 A 632 grade TP347	6/ 0 10 30 50 65		1.006 1.011 1.500 3.040 9.243	2.3 - 2.5	2	Slightly magnetic after heavy cold work Class 2 in accordance with MIL-STD-288
	Cyclops 19-9-DL					25	1.04			1(m)	

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TABLE IV. Wrought austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Chromium-nickel-iron alloy	Timken 16-25-6		0.12 C 2.0 Mn 1.0 Si 15.0 - 17.5 Cr 24 - 27 Ni 5.5 - 7 Mo 0.07 N rem Fe			6/ 25	1.02			1(m)	
Chromium-nickel-iron alloy	Discaloy	S66200	0.08 max C 12 - 15 Cr 24 - 28 Ni 2.5 - 3.5 Mo 1.3 - 1.8 Ti 0.6 Mn rem Al		ASTM A 453 A 564 A 638 B 637 B 639	UNK	1.02			1(m)	
	Therm-alloy 30						1.00			1(m)	
	Thompson alloy		0.4 - 0.5 C 0.7 max Mn 13 - 15 Cr 0.5 Mo 1.75 - 3.0 W rem Fe				1.11			1(m)	

1/ Unless otherwise indicated, values shown are maximum permitted.
2/ Percent IACS calculated on a volume basis in accordance with ASTM B 193 from data provided in the indicated data sources.
3/ See 3.1.8(i) for explanation of code.
4/ Specifies optional ordering of low permeability steel.
5/ Specifies material to have a permeability not to exceed 1.2 in the annealed condition.
6/ Two or more sets of magnetic permeability data for a single type of steel are provided to show the effects of cold-working.

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TABLE V. Precipitation hardening corrosion resistant steels.

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Steel, corrosion resistant, precipitation hardening	17-4 PH	S63000	0.07 C 1.0 Mn 0.04 P 0.03 S 15.5 - 17.5 Cr 3.0 - 5.0 Ni 0.15-0.45 Nb+Ta 3.0 - 5.0 Cu 1.0 Si rem Fe	MIL-C-24111 MIL-S-81591	ASTM A 564 A 638 B 637 B 639	0	29.5			1(m)	
Steel, corrosion resistant, precipitation hardening	17-7 PH		0.09 C 1.0 Mn 0.04 P 0.03 S 16.0 - 18.0 Cr 6.50 - 7.75 Ni 0.75 - 1.5 Al 1.0 Si rem Fe	MIL-S-25043		0	21.5			1(m)	
Steel, corrosion resistant, precipitation hardening	Type 322					0	39.0			1(m)	

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TABLE V. Precipitation hardening corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
	Crucible 3003		0.60 C 4.5 Mn 15.0 Ni 12.5 Cr 3.0 Al rem Fe			UNK	1.02-1.62			1(m)	
	Rezistal 3311					UNK	1.02			1(m)	
	Rezistal 3329					UNK	1.02-14.2			1(m)	
	Armco 21-4					UNK	1.02			1(m)	
Special chromium-nickel-phosphorous alloy			0.15 C 1.00 Mn 0.20 - 0.40 P 0.040 S 1.00 Si 9.5 - 12.0 Ni 16.0 - 18.0 Cr	MIL-S-17759 4/		UNK	1.02			1(m)	

See footnotes at end of table.

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TABLE V. Precipitation hardening corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Corrosion resistant (stainless) steel	Type 329	S32900	0.10 C 2.00 Mn 1.00 Si 25.0 - 30.00 Cr 3.00 - 6.00 Ni 1.00 - 2.00 Mo 0.040 P 0.030 S			UNK	8.00		2.3	1(m) 6(e)	

1/ Unless otherwise indicated, values shown are maximum permitted.
2/ Percent IACS calculated on a volume basis in accordance with ASTM B 193 from data provided in the indicated data sources.
3/ See 3.1.8(i) for explanation of code.
4/ Specifies material to have a permeability not to exceed 1.2.

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TABLE VI. Cast austenitic corrosion resistant steels.

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Cast corrosion resistant steel	Type 304	J92610	18.0 - 20.0 Cr 8.0 - 12.0 Ni 0.08 C 2.00 Mn 0.04 P 0.03 S 1.00 Si	MIL-S-81591		0	1.07		2.5	1(m)	
Cast corrosion resistant steel	Type 347	J92640	0.08 C 17.0 - 19.5 Cr 2.00 Mn 9.0 - 13.0 Ni 0.04 P 0.03 S 1.00 Si 10 x C min Nb+Ta	MIL-S-81591		0	1.03			1(m)	
Cast corrosion resistant steel	Type 316	J92810	16.0 - 18.0 Cr 10.0 - 14.0 Ni 2.0 - 3.0 Mo 0.08 C 2.00 Mn 0.04 P 0.03 S 1.00 Si	MIL-S-81591		0	1.43		2.5	1(m)	
Cast corrosion resistant steel	18-8 Cb					0	2.14			1(m)	

See footnotes at end of table.

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TABLE VI. Cast austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Heat-resistant iron-chromium-nickel alloy casting	25-20 AMS 5366	J94211	0.20 - 0.60 C 2.00 Mn 2.00 Si 0.04 P 0.04 S 24.0 - 28.0 Cr 18.0 - 22.0 Ni 0.50 Mo		ASTM A 297 grade HK	0	1.02			1(m)	
Cast iron	Ductile Ni-resist					0	1.03			1(m)	
Cast iron	Ni-resist					UNK	1.03			1(m)	
Cast iron	Ductile Ni-resist type 2		18 - 22 Ni 2.0 - 4.0 Cr 2.0 - 3.0 C 1.0 - 1.5 Mn 1.0 - 2.5 Si rem Fe			UNK	1.03		1.01	1(m) 7(e)	
Welding rod	19-9 weld Chromend K		0.07 C 19.0 Cr 9.0 Ni rem Fe			0	1.15			1(m)	
Weld 18-8 corrosion resistant steel	18-8 Cb weld Stainweld A-5 Cb					0	2.06			1(m)	

See footnotes at end of table.

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TABLE VI. Cast austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Weld	19-9 Cb weld, long. weld					0	1.77			1(m)	
Weld	19-9 Cb weld, trans. weld					0		1.04		1(m)	
Weld	19-9 Ti weld					0	1.30			1(m)	
Welding electrode	18-13-2 weld Chromend KMO		0.08 C 2.0 Mo 18.0 Cr 13.0 Ni rem Fe			0	1.05			1(m)	
Weld metal	25-12 weld Chromend H.C.		0.10 C 25.0 Cr 12.0 Ni rem Fe			0	1.57			1(m)	
Weld	25-12 Ti weld					0	1.07			1(m)	
Weld	25-20 weld Chromend H.C.N.					0	1.00			1(m)	

See footnotes at end of table.

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TABLE VI. Cast austenitic corrosion resistant steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Weld	25-20		0.10 C 25.0 Cr 20.0 Ni rem Fe			0	1.02			1(m)	
Weld	Stainless 304 and weld metal 19:9 + 308					0	1.07			1(m)	
Weld	Stainless 304 and weld metal 19:9 + 308					0	1.10			1(m)	
Weld	Stainless 304 and weld metal 19:9 + 308					0	1.07			1(m)	
Welding electrodes	Frogalloy M		0.5 - 0.6 C 4.0 min Mn 18.0 - 21.0 Cr 9.0 - 10.5 Ni rem Fe			0		1.00		1(m)	

1/ Unless otherwise indicated, values shown are maximum permitted.
2/ Percent IACS calculated on a volume basis in accordance with ASTM B 193 from data provided in the indicated data sources.
3/ See 3.1.8(i) for explanation of code.

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TABLE VII. Austenitic manganese steels.

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
Hadfield manganese steel			1.0 - 1.35 C 10.5 - 15.0 Mn 0.10 max P 0.1 - 0.30 Si rem Fe			0 26	1.02 1.02		1.2	1(m) 7(e)	
Manganese steel	Jessup no. 9		0.40 C 13.0 Mn 3.5 Ni 4.0 Cr			UNK 31.1	1.02 1.08			1(m)	
Manganese-nickel steel			0.70 C 14.0 Mn 3.0 Ni			0	1.01			1(m)	
Manganese-nickel steel			0.80 C 14.0 Mn 4.0 Ni			0	1.00			1(m)	
Manganese steel	Jessup no. 200		0.30 C 11.0 Mn 7.0 Ni			11	1.02			1(m)	
Manganese-nickel steel			0.3 C 11.0 Mn 8.0 Ni		ASTM A 289	0	1.00			1(m)	

See footnotes at end of table.

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TABLE VII. Austenitic manganese steels. - Continued

Common name	Commercial designation	UNS No.	Chemical composition (percent)1/	Specifications		Percent cold-work	Magnetic permeability		Electrical conductivity (% IACS)2/	Data sources3/	Remarks
				Government	Commercial		Norm	Max			
-	Chromanal		1.1 C 12.0 Mn 2.0 Cr			UNK	1.31			1(m)	
Manganese steel	Midvale nonmagnetic		0.70 C 8.0 Mn 9.0 Ni 4.0 Cr				1.00				

1/ Unless otherwise indicated, values shown are maximum permitted.
2/ Percent IACS calculated on a volume basis in accordance with ASTM B 193 from data provided in the indicated data sources.
3/ See 3.1.8(i) for explanation of code.

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TABLE VIII. Data sources.

Numeric code	Type of data electrical (e)/magnetic (m)	Source document
1	(m)	E.E.S. (Engineering Experiment Station; predecessor to NSRDC - Annapolis) Report 4E(2)66904, 4P(2)66918, (Magnetic Characteristics of "Non-Magnetic" Metallic Materials, Comparison of Properties in Strong and Weak Fields.)
2	(e) (m)	Material and chemical analysis provided by ARMCO. All of the materials investigated, except the type 310 heat treatment were annealed at 1066°C and air cooled. The type 310 heat treatment was annealed at 1149°C and air cooled.
3	(e)	MIL-HDBK-694.
4	(e)	MIL-HDBK-698.
5	(e)	National Bureau of Standards Circular 592, (Nickel and Its Alloys).
6	(e)	American Iron and Steel Institute, (Steel Products Manual, Stainless and Heat Resisting Steels).
7	(e)	Metals Reference Book, fifth edition.
8	(e)	SAE J470.

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TABLE IX. Cross-index - Government to commercial specifications.

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
QQ-A-596	ASTM B 26 B 108	A03550	5 percent silicon-copper-magnesium alloy	Aluminum
QQ-A-596	ASTM B 26		213	Aluminum
QQ-A-601	ASTM B 26 B 108	A03550	5 percent silicon-copper-magnesium alloy	Aluminum
QQ-B-626	ASTM B 16	C36000	SAE CA360	Brass, free cutting
QQ-B-637	ASTM B 21 B 124 B 174 B 283	C46400	SAE CA464	Naval brass, uninhibited
QQ-B-639	ASTM B 21 B 124 B 174 B 283	C46400	SAE CA464	Naval brass, uninhibited
QQ-B-728	ASTM B 124 B 138 B 283	C67500	Manganese bronze A	Manganese, bronze wrought
QQ-C-390	ASTM B 30 B 505 B 584	C94700	---	Tin nickel bronze
QQ-C-390	ASTM B 505 B 584	C94800	---	Tin nickel bronze
QQ-C-390	ASTM B 148	C95400	Grade C	Aluminum bronze
QQ-C-390	ASTM B 30 B 148	C95700	---	Aluminum manganese bronze
QQ-C-390	ASTM B 30 B 148 B 271 B 505	C95800	---	Aluminum nickel bronze

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
QQ-C-390	ASTM B 148	C95200	Grade A AMPCO 12	Aluminum bronze
QQ-C-390	ASTM B 30 B 271 B 584	C87400	Silicon brass	Copper-silicon alloy
QQ-C-465	ASTM B 150 B 283 B 171 B 124	C63000	AMPCO 45	Aluminum nickel bronze
QQ-C-591	ASTM B 96 B 98 B 100 B 124 B 283 B 315	C65500	High silicon bronze A	Copper-silicon alloy
QQ-N-281	ASTM F 467 F 468	N04405	Class B (Monel alloy R-405)	Nickel-copper-alloy (R-Monel)
QQ-N-286	---	N05502	Monel alloy 502	Nickel-copper-aluminum alloy (K-Monel)
QQ-N-288	---	---	---	Nickel-copper-silicon alloy (S-Monel)
QQ-S-763	ASTM A 276 A 412 A 666	S20200	Type 202	Chromium-manganese nickel corrosion resistant (stainless) steel
QQ-S-763	ASTM A 167 A 240 A 276 A 479 A 493 A 511 A 554 A 666	S30200	Type 302	Chromium-nickel-corrosion resistant (stainless) steel

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
QQ-S-763	ASTM A 213 A 240 A 249 A 262 A 269 A 276 A 312 A 403 A 479 A 493 A 498 A 511 A 554 A 632 A 666 A 688	S30400	Type 304	Corrosion resistant (stainless) steel
QQ-S-763	ASTM A 167 A 249 A 276 A 493 A 498 A 511 A 554	S30500	Type 305	Corrosion resistant (stainless) steel
QQ-S-763	ASTM A 249 A 276 A 312 A 403 A 498 A 511 A 554	S30900	Type 309	Corrosion resistant (stainless) steel
QQ-S-763	ASTM A 167 A 213 A 249 A 276 A 312 A 403 A 498 A 511 A 554 A 632	S31000	Type 310	Corrosion resistant (stainless) steel

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
QQ-S-763	ASTM A 167 A 213 A 240 A 249 A 262 A 269 A 276 A 312 A 403 A 479 A 493 A 498 A 511 A 554 A 632 A 651 A 666 A 688	S31600	Type 316	Corrosion resistant (stainless) steel
QQ-S-763	ASTM A 167 A 213 A 240 A 249 A 262 A 269 A 276 A 312 A 403 A 479 A 498 A 511 A 554 A 632	S32100	Type 321	Corrosion resistant (stainless) steel

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
QQ-S-763	ASTM A 167 A 213 A 240 A 249 A 262 A 269 A 312 A 403 A 479 A 493 A 498 A 511 A 554 A 632	S34700	Type 347	Corrosion resistant (stainless) steel
QQ-S-766	ASTM A 276 A 412 A 666	S20100	Type 201	Chromium-manganese nickel corrosion resistant (stainless) steel
QQ-S-766	ASTM A 276 A 412 A 666	S20200	Type 202	Chromium-manganese-nickel corrosion resistant (stainless) steel
QQ-S-766	ASTM A 167 A 177 A 554	S30100	Type 301	Chromium-nickel-corrosion-resistant (stainless) steel
QQ-S-766	ASTM A 167 A 240 A 276 A 479 A 493 A 511 A 554 A 666	S30200	Type 302	Chromium-nickel corrosion resistant (stainless) steel

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
QQ-S-766	ASTM A 213 A 240 A 249 A 262 A 269 A 276 A 312 A 403 A 479 A 493 A 498 A 511 A 554 A 632 A 666 A 688	S30400	Type 304	Corrosion resistant (stainless) steel
QQ-S-766	ASTM A 167 A 249 A 276 A 493 A 498 A 511 A 554	S30500	Type 305	Corrosion resistant (stainless) steel
QQ-S-766	ASTM A 249 A 276 A 312 A 403 A 498 A 511 A 554	S30900	Type 309	Corrosion resistant (stainless) steel
QQ-S-766	ASTM A 167 A 213 A 249 A 276 A 312 A 403 A 498 A 511 A 554 A 632	S31000	Type 310	Corrosion resistant (stainless) steel

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
QQ-S-766	ASTM A 167 A 213 A 240 A 249 A 262 A 269 A 276 A 312 A 403 A 479 A 493 A 498 A 511 A 554 A 632 A 651 A 666 A 688	S31600	Type 316	Corrosion resistant (stainless) steel
QQ-S-766	ASTM A 167 A 213 A 240 A 249 A 262 A 269 A 276 A 312 A 403 A 479 A 498 A 511 A 554 A 632	S32100	Type 321	Corrosion resistant (stainless) steel
QQ-S-766	ASTM A 167 A 213 A 240 A 249 A 262 A 269 A 312 A 403 A 479 A 493 A 498 A 511 A 554 A 632	S34700	Type 347	Corrosion resistant (stainless) steel

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
QQ-W-423	ASTM A 167 A 240 A 276 A 479 A 493 A 511 A 554 A 666	S30200	Type 302	Corrosion resistant (stainless) steel
MIL-S-862	ASTM A 167 A 240 A 276 A 479 A 493 A 511 A 554 A 666	S30200	Type 302	Chromium-nickel corrosion resistant (stainless) steel
MIL-S-5059	ASTM A 167 A 177 A 554	S30100	Type 301	Chromium-nickel corrosion resistant (stainless) steel
MIL-T-5695	ASTM A 167 A 177 A 554	S30100	Type 301	Chromium-nickel corrosion-resistant (stainless) steel
MIL-W-6712	ASTM A 167 A 240 A 276 A 479 A 493 A 511 A 554 A 666	S30200	Type 302	Chromium-nickel corrosion-resistant (stainless) steel
MIL-T-8231	ASTM B 96 B 98 B 100 B 124 B 283 B 315	C65500	High silicon bronze A	Copper silicon alloy
MIL-C-11866	ASTM B 30 B 271 B 505 B 584	C86200	High-strength manganese bronze SAE CA862	High-strength yellow brass

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
MIL-C-11866	ASTM B 26 B 108	A03550	5 percent silicon-copper-manganese	Aluminum
MIL-T-15005	ASTM B 111 B 121 B 122 B 151 B 171 B 359 B 395 B 402 B 466 B 467 B 543 B 552 B 608	C71500	---	70-30 copper-nickel alloy
MIL-C-15345	ASTM B 30 B 61 B 271 B 505 B 584	C92200	Steam or valve bronze-Navy "M" SAE CA922	Leaded tin bronze
MIL-C-15345	ASTM B 30 B 271 B 505 B 584	C90300	88-8-0-4 or modified tin SAE CA903	Tin bronze
MIL-C-15345	ASTM B 30 B 271 B 584	C86500	No. 1 manganese SAE CA43	Manganese bronze
MIL-C-15345	ASTM B 30 B 148 B 271 B 505	C95800	---	Aluminum nickel bronze
MIL-C-15345	ASTM B 30 B 271 B 505 B 584	C83600	85-5-5-5 or no. 1 composition SAE CA836	Red brass, leaded

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
MIL-C-15726	ASTM B 111 B 121 B 122 B 151 B 171 B 359 B 395 B 402 B 466 B 467 B 543 B 552 B 608	C71500	---	70-30 copper nickel alloy
MIL-B-16166	ASTM B 124 B 150 B 171 B 283	C63000	AMPCO 45	Aluminum nickel bronze
MIL-T-16420	ASTM B 111 B 121 B 122 B 151 B 171 B 359 B 395 B 402 B 466 B 467 B 543 B 552 B 608	C71500	---	70-30 copper-nickel alloy
MIL-N-17163	---	---	Monel 402 (formerly 326 monel)	Monel
MIL-S-17759	---	---	---	Special chromium-nickel-phosphorus alloy
MIL-S-17996	ASTM A 276 A 412 A 666	S20100	Type 201	Chromium-manganese nickel corrosion resistant (stainless) steel

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
MIL-B-21230	ASTM B 30 B 148 B 271 B 505	C95800	---	Aluminum nickel bronze
MIL-C-22087	ASTM B 30 B 271 B 584	C86500	No. 1 Manganese SAE CA43	Manganese bronze
MIL-C-22087	ASTM B 30 B 271 B 505 B 584	C83600	85-5-5-5 or no. 1 composition SAE CA836	Red brass, leaded
MIL-C-22229	ASTM B 30 B 271 B 505 B 584	C86200	High-strength manganese bronze SAE CA862	High-strength yellow brass
MIL-C-22229	ASTM B 30 B 271 B 505 B 584	C90300	88-8-0-4 or modified tin SAE CA903	Tin bronze
MIL-C-22229	ASTM B 30 B 271 B 584	C86500	No. 1 manganese CA43	Manganese bronze
MIL-C-22229	ASTM B 30 B 271 B 505 B 584	C83600	85-5-5-5 or no. 1 composition SAE CA836	Red brass, leaded
MIL-C-24111	ASTM A 564 A 638 B 637 B 639	S63000	17-4 PH	Steel, corrosion-resistant, precipitation hardening
MIL-S-25043	---	---	17-7 PH	Steel, corrosion-resistant, precipitation hardening

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TABLE IX. Cross-index - Government to commercial specifications. - Continued

Government specification	Commercial specification	UNS no.	Commercial designation	Common name
MIL-S-81591	ASTM A 564 A 638 B 637 B 639	S63000	17-4 PH	Steel corrosion-resistant, precipitation hardening
MIL-S-81591		J92610 J92640 J92810 J94211	Type 304 Type 347 Type 316 25-20 AMS 5366	Cast corrosion resistant steel Cast corrosion resistant steel Cast corrosion resistant steel Heat-resistant iron-chromium-nickel alloy casting

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TABLE X. Cross-index - commercial to Government specifications.

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM A 167	QQ-S-766 MIL-S-5059 MIL-T-5695	S30100	Type 301	Chromium-nickel-corrosion-resistant (stainless) steel
A 167	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712	S30200	Type 302	Chromium-nickel-corrosion resistant (stainless) steel
A 167	QQ-S-763 QQ-S-766	S30500	Type 305	Corrosion resistant (stainless) steel
A 167	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 167	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 167	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 167	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 177	QQ-S-766 MIL-S-5059 MIL-T-5695	S30100	Type 301	Chromium-nickel-corrosion resistant (stainless) steel
A 213	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 213	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 213	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 213	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 213	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM A 240	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712	S30200	Type 302	Chromium-nickel-corrosion resistant (stainless) steel
A 240	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 240	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 240	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 240	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 249	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 249	QQ-S-763 QQ-S-766	S30500	Type 305	Corrosion resistant (stainless) steel
A 249	QQ-S-763 QQ-S-766	S30900	Type 309	Corrosion resistant (stainless) steel
A 249	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 249	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 249	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 249	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 262	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 262	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 262	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM A 262	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 269	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 269	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 269	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 269	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 276	QQ-S-766 MIL-S-17996	S20100	Type 201	Chromium-manganese-nickel corrosion resistant (stainless) steel
A 276	QQ-S-763 QQ-S-766	S20200	Type 202	Chromium-manganese-nickel corrosion resistant (stainless) steel
A 276	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712	S30200	Type 302	Chromium-nickel-corrosion resistant (stainless) steel
A 276	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 276	QQ-S-763 QQ-S-766	S30500	Type 305	Corrosion resistant (stainless) steel
A 276	QQ-S-763 QQ-S-766	S30900	Type 309	Corrosion resistant (stainless) steel
A 276	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 276	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM A 276	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 297	---	J94211	25-20 AMS 5366	Heat-resistant iron-chromium-nickel alloy casting
A 312	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 312	QQ-S-763 QQ-S-766	S30900	Type 309	Corrosion resistant (stainless) steel
A 312	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 312	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 312	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 312	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 403	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 403	QQ-S-763 QQ-S-766	S30900	Type 309	Corrosion resistant (stainless) steel
A 403	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 403	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 403	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 403	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM A 412	QQ-S-766 MIL-S-17996	S20100	Type 201	Chromium-manganese-nickel corrosion resistant (stainless) steel
A 412	QQ-S-763 QQ-S-766	S20200	Type 202	Chromium-manganese-nickel corrosion resistant (stainless) steel
A 479	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712	S30200	Type 302	Chromium-nickel-corrosion resistant (stainless) steel
A 479	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 479	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 479	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 479	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 493	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712	S30200	Type 302	Corrosion resistant (stainless) steel
A 493	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 493	QQ-S-763 QQ-S-766	S30500	Type 305	Corrosion resistant (stainless) steel
A 493	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 493	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM A 498	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 498	QQ-S-763 QQ-S-766	S30500	Type 305	Corrosion resistant (stainless) steel
A 498	QQ-S-763 QQ-S-766	S30900	Type 309	Corrosion resistant (stainless) steel
A 498	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 498	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 498	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 498	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 511	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712	S30200	Type 302	Chromium-nickel-corrosion resisting (stainless) steel
A 511	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 511	QQ-S-763 QQ-S-766	S30500	Type 305	Corrosion resistant (stainless) steel
A 511	QQ-S-763 QQ-S-766	S30900	Type 309	Corrosion resistant (stainless) steel
A 511	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 511	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 511	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM A 511	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 554	QQ-S-766 MIL-S-5059 MIL-T-5695	S30100	Type 301	Chromium-nickel-corrosion resistant (stainless) steel
A 554	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712	S30200	Type 302	Chromium-nickel-corrosion resistant (stainless) steel
A 554	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 554	QQ-S-763 QQ-S-766	S30500	Type 305	Corrosion resistant (stainless) steel
A 554	QQ-S-763 QQ-S-766	S30900	Type 309	Corrosion resistant (stainless) steel
A 554	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel
A 554	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 554	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 554	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 564	MIL-C-24111 MIL-S-81591	S63000	17-4 PH	Steel, corrosion-resistant, precipitation hardening
A 632	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 632	QQ-S-763 QQ-S-766	S31000	Type 310	Corrosion resistant (stainless) steel

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM A 632	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 632	QQ-S-763 QQ-S-766	S32100	Type 321	Corrosion resistant (stainless) steel
A 632	QQ-S-763 QQ-S-766	S34700	Type 347	Corrosion resistant (stainless) steel
A 638	MIL-C-24111 MIL-S-81591	S63000	17-4 PH	Steel, corrosion-resistant, precipitation hardening
A 651	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 666	QQ-S-766 MIL-S-17996	S20100	Type 201	Chromium-manganese-nickel corrosion resistant (stainless) steel
A 666	QQ-S-763 QQ-S-766	S20200	Type 202	Chromium-manganese-nickel corrosion resistant (stainless) steel
A 666	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712	S30200	Type 302	Chromium-nickel-corrosion resistant (stainless) steel
A 666	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 666	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel
A 688	QQ-S-763 QQ-S-766	S30400	Type 304	Corrosion resistant (stainless) steel
A 688	QQ-S-763 QQ-S-766	S31600	Type 316	Corrosion resistant (stainless) steel

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM B 16	QQ-B-626	C36000	SAE CA360	Brass, free cutting
B 21	QQ-B-637 QQ-B-639	C46400	SAE CA464	Naval brass, uninhibited
B 26	QQ-A-596 QQ-A-601 MIL-C-11866	A03550	5 percent silicon-copper copper-manganese alloy AA355.0	Aluminum
	QQ-A-596		213	Aluminum
	MIL-C-15345	C92200	Steam or valve bronze - Navy "M" SAE CA922	Leaded tin bronze
	QQ-C-390	C94700	---	Tin nickel bronze
	MIL-C-11866 MIL-C-22229	C86200	High-strength manganese bronze SAE CA862	High-strength yellow brass
	MIL-C-15345 MIL-C-22229	C90300	88-8-0-4 or modified tin SAE CA903	Tin bronze
	MIL-C-15345 MIL-C-22087 MIL-C-22229	C86500	No. 1 manganese SAE CA43	Manganese bronze
	QQ-C-390	C95700	---	Aluminum manganese bronze
	QQ-C-390 MIL-C-15345 MIL-B-21230	C95800	---	Aluminum nickel bronze
	MIL-C-22229 MIL-C-22087 MIL-C-15345	C83600	85-5-5-5 or no. 1 composition SAE CA836	Red brass, leaded

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
	QQ-C-390	C87400	Silicon brass	Copper-silicon alloy
ASTM B 61	MIL-C-15345	C92200	Steam or valve bronze-Navy "M" SAE CA922	Leaded tin bronze
B 96	QQ-C-591 MIL-T-8231	C65500	High silicon bronze A	Copper-silicon alloy
B 98	QQ-C-591 MIL-T-8231	C65500	High silicon bronze A	Copper-silicon alloy
B 100	QQ-C-591 MIL-T-8231	C65500	High silicon	Copper-silicon alloy
B 108	QQ-A-596 QQ-A-601 MIL-C-11866	A03550	5 percent silicon-copper-manganese alloy	Aluminum
B 111	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500		70-30 copper-nickel alloy
B 111	---	C71000	---	80-20 copper-nickel alloy
B 121	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper-nickel alloy
B 122	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper-nickel alloy
B 122	---	C71000	---	80-20 copper nickel alloy
B 124	QQ-C-465 MIL-B-16166	C63000	AMPCO 45	Aluminum nickel bronze
B 124	QQ-C-591 MIL-T-8231	C65500	High silicon bronze A	Copper-silicon alloy
B 124	QQ-B-728	C67500	Manganese bronze A	Manganese bronze, wrought

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM B 124	QQ-B-637 QQ-B-639	C46400	SAE CA464	Naval brass, uninhibited
B 138	QQ-B-728	C67500	Manganese bronze A	Manganese bronze, wrought
B 148	QQ-C-390	C95400	Grade C	Aluminum bronze
B 148	QQ-C-390	C95700	---	Aluminum manganese bronze
B 148	QQ-C-390 MIL-C-15345 MIL-B-21230	C95800	---	Aluminum nickel bronze
B 148	QQ-C-390	C95200	Grade A AMPCO 12	Aluminum bronze
B 150	QQ-C-465 MIL-B-16166	C63000	AMPCO 45	Aluminum nickel bronze
B 151	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper-nickel alloy
B 164	QQ-N-281	N04405	Class B (Monel alloy R-405)	Nickel-copper-alloy (R-Monel)
B 171	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper nickel alloy
B 171	QQ-C-465 MIL-B-16166	C63000	AMPCO 45	Aluminum nickel bronze
B 174	QQ-B-637 QQ-B-639	C46400	SAE CA464	Naval brass, uninhibited
B 206	---	C71000	---	80-20 copper-nickel alloy
B 271	MIL-C-15345	C92200	Steam or valve bronze-Navy "M" SAE CA922	Leaded tin bronze

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM B 271	MIL-C-11866 MIL-C-22229	C86200	High-strength manganese bronze SAE CA862	High-strength yellow brass
B 271	MIL-C-15345 MIL-C-22229	C90300	88-8-0-4 or modified tin SAE CA903	Tin bronze
B 271	MIL-C-15345 MIL-C-22087 MIL-C-22229	C86500	No. 1 manganese SAE CA865	Manganese bronze
B 271	QQ-C-390 MIL-C-15345 MIL-B-21230	C95800	---	Aluminum nickel bronze
B 271	MIL-C-22229 MIL-C-22087 MIL-C-15345	C83600	85-5-5-5 or no. 1 composition SAE CA836	Red brass, leaded
B 271	QQ-C-390	C87400	Silicon brass	Copper-silicon alloy
B 283	QQ-C-591 MIL-T-8231	C65500	High silicon bronze A	Copper-silicon alloy
B 283	QQ-B-728	C67500	Manganese bronze A	Manganese bronze, wrought
B 283	QQ-C-465 MIL-B-16166	C63000	AMPCO 45	Aluminum nickel bronze
B 283	QQ-B-637 QQ-B-639	C46400	SAE CA464	Naval brass, uninhibited
B 315	QQ-C-591 MIL-T-8231	C65500	High silicon bronze A	Copper-silicon alloy
B 359	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper-nickel alloy
B 395	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper-nickel alloy

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM B 402	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper-nickel alloy
B 466	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper-nickel alloy
B 467	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper-nickel alloy
B 505	QQ-C-390	C94700	---	Tin nickel bronze
B 505	QQ-C-390	C94800	---	Tin nickel bronze
B 505	MIL-C-11866 MIL-C-22229	C86200	High-strength manganese bronze SAE CA862	High-strength yellow brass
B 505	MIL-C-22229	C90300	88-8-0-4 or modified tin SAE CA903	Tin bronze
B 505	QQ-C-390 MIL-B-21230	C95800	---	Aluminum nickel bronze
B 505	MIL-C-22229 MIL-C-22087	C83600	85-5-5-5 or no. 1 composition SAE CA836	Red brass, leaded
B 543	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper nickel alloy
B 552	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper nickel alloy
B 584	MIL-C-15345	C92200	Steam or valve bronze-Navy "M" SAE CA922	Leaded tin bronze

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TABLE X. Cross-index - commercial to Government specifications. - Continued

Commercial specification	Government specification	UNS no.	Commercial designation	Common name
ASTM B 584	QQ-C-390	C94700	---	Tin nickel bronze
B 584	QQ-C-390	C94800	---	Tin nickel bronze
B 584	MIL-C-11866 MIL-C-22229	C86200	High-strength manganese bronze SAE CA862	High-strength, yellow brass
B 584	MIL-C-15345 MIL-C-22229	C90300	88-8-0-4 or modified tin SAE CA903	Tin bronze
B 584	MIL-C-15345 MIL-C-22087 MIL-C-22229	C86500	No. 1 manganese SAE CA43	Manganese bronze
B 584	MIL-C-22229 MIL-C-22087 MIL-C-15345	C83600	85-5-5-5 or no. 1 composition SAE CA836	Red brass leaded
B 584	QQ-C-390	C87400	Silicon brass	Copper-silicon alloy
B 608	MIL-T-15005 MIL-C-15726 MIL-T-16420	C71500	---	70-30 copper nickel alloy
B 637	MIL-C-24111 MIL-S-81591	S63000	17-4 PH	Steel, corrosion resistant, precipitation hardening
B 639	MIL-C-24111 MIL-S-81591	S63000	17-4 PH	Steel, corrosion resistant, precipitation hardening
F 467	QQ-N-281	N04405	Class B (Monel alloy R-405)	Nickel-copper-alloy (R-Monel)
F 468	QQ-N-281	N04405	Class B (Monel alloy R-405)	Nickel-copper-alloy (R-Monel)

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TABLE XI. Cross-index - UNS numbers to Government specifications.1/

UNS no.	Government specification	UNS no.	Government specification
A03550	QQ-A-596 QQ-A-601 MIL-C-11866	C90300	MIL-C-15345 MIL-C-22229
C36000	QQ-B-626	C92200	MIL-C-15345
C46400	QQ-B-637 QQ-B-639	C94700	QQ-C-390
		C94800	QQ-C-390
C63000	QQ-C-465 MIL-B-16166	C95200	QQ-C-390
C65500	QQ-C-591 MIL-T-8231	C95400	QQ-C-390
C67500	QQ-B-728	C95700	QQ-C-390
C7100	---	C95800	QQ-C-390 MIL-C-15345 MIL-B-21230
C71500	MIL-T-15005 MIL-C-15726 MIL-T-16420	J92610	MIL-S-81591
C83600	MIL-C-22229 MIL-C-22087 MIL-C-15345	J92640	MIL-S-81591
C86200	MIL-C-11866 MIL-C-22229	J92810	MIL-S-81591
		J94211	---
C86500	MIL-C-15345 MIL-C-22087 MIL-C-22229	N04405	QQ-N-281 QQ-N-286
		N05502	QQ-N-286
C87400	QQ-C-390	S20100	QQ-S-766 MIL-S-17996
S20200	QQ-S-763 QQ-S-766	S63000	MIL-C-24111 MIL-S-81506 MIL-S-81591

See footnote at end of table.

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TABLE XI. Cross-index - UNS numbers to Government specifications.[1/] - Continued

UNS no.	Government specification	UNS no.	Government specification
S20500	---		
S30100	QQ-S-766 MIL-S-5059 MIL-T-5695		
S30200	QQ-S-763 QQ-S-766 QQ-W-423 MIL-S-862 MIL-W-6712		
S30300	---		
S30400	QQ-S-763 QQ-S-766		
S30500	QQ-S-763 QQ-S-766		
S30900	QQ-S-763 QQ-S-766		
S31000	QQ-S-763 QQ-S-766		
S31600	QQ-S-763 QQ-S-766		
S32100	QQ-S-763 QQ-S-766		
S34700	QQ-S-763 QQ-S-766		

[1/] See Government specification in table IX for cross-index to commercial specification, commercial designation, and common name.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-HDBK-270(SH)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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